

B30801 Managerial Accounting Autumn 2025

Doug Skinner

**Session #1: Rules of the Game; Costs;
Contribution Margins and CVP Analysis.**

CHICAGO BOOTH



The University of Chicago Booth School of Business

Managerial Accounting

Getting the right information for:

- Strategic decisions: evaluating new products/ventures, pricing, product design, product mix, manufacturing.
- Performance evaluation and compensation.
- Coordination and Control.

Non-financial measures – such as customer satisfaction and employee loyalty – also important (includes ESG/sustainability).

Different decisions require different information.

Connects to Micro; Corporate Finance; Ops; Marketing; Stats.

Some examples:

- How quickly will my start-up become profitable and generate cash flow? *See “Private jet start-ups hit an air pocket.” FT 8.2.25.*
- Will overhaul of new plant create value? *See “Ford to spend \$2B overhauling Kentucky plant in push for cheaper EVs” (priced from \$30K). FT 8.11.25.*
- Should I invest in battery-swapping infrastructure? (CATL). *See “Will a 3-minute battery swap beat a 5-minute charge for China’s EV drivers?” FT 8.18.25.*

	Managerial, Cost Accounting	Financial Accounting
Purpose	Management tool for internal , data-driven decision-making; performance evaluation; control.	Provides information to external users, primarily for valuation, governance.
Orientation	Internal. Future decisions, execution. Emphasis on decision relevance. Internal users at all levels.	External. Historical: reports on past transactions and events. Emphasis on reliable numbers.
Rules	No rules! Consider cost-benefit of data for decisions.	US GAAP/IFRS rules as required by regulation.
Type of data/ information	Cost as defined by economists (not financial accountants). Includes non-financial and qualitative information. Forward- and backward looking.	Historical financial data prepared for past periods (quarterly, annual). Costs: either product costs or period costs. Costs vs. Expenses.
Personality type	Relaxed, friendly, happy.	Not so much.

Consider a large tech company:

- Multiple reportable segments (core business; cloud services; “moonshots”).
- Within each segment, different platforms and ecosystems (built around operating system, such as iOS).
- Within each platform/ecosystem, multiple product groups.
 - Within each product group, multiple products.
- Almost none of this is visible in 10-K.

Uses of internal P&Ls (by board/C-suite and down):

- Measure performance; evaluate; compensate.
- Control.
- Evaluate teams;
- Evaluate products – resource allocation;
- Communicate strategic priorities;
- Product managers manage teams/businesses.
- Produce numbers for GAAP roll-up.

Each purpose somewhat different and may require different P&Ls or versions thereof (different levels of aggregation, different cost allocations)

Session #1	Cost concepts. Contribution margin/CVP analysis
Session #2	Cost allocation. Absorption costing. Knowledge in orgns.
Session #3	Why allocate costs? Activity-based Costing (ABC)
Session #4	Responsibility Centers. Budgets. Transfer Pricing
Session #5	Non-financial Performance Measurement and Balanced Scorecard

Key resources (all on Canvas):

- PPT decks/class discussion.
- Case decks/solutions (including FinePrint).
- Lecture examples (ZPhone, Bullwinkle, Variance Analysis Questions).
- Readings (some optional).
- Past exams/solutions

Case	Topic
RegionFly	Indirect Cost Allocation: Supply-driven “death spiral”
VMD MIC	Indirect Cost Allocation: Demand-driven “death spiral”
Midwest Office Products	Time-driven Activity-Based Costing (ABC)
Birch Paper	Transfer Pricing
Citibank Performance Evaluation	Non-financial performance measurement and Balanced Scorecard

Convey *both* quantitative tools + how to use them to make good decisions that require management judgment.

What's the right objective?

Where do we start?

- We manage an organization, and want that organization to run more effectively and efficiently—to better achieve its goals.
- For business firms, particularly publicly-traded companies, managers' goal is to *maximize long-term value* of the firm—the wealth of its owners.

Where do we start?

- Jensen argues that organizations need a **single-valued** objective function, as opposed to multiple objectives.
- Why? Managers *need a single-dimensioned scorecard* to evaluate performance and decide between different courses of action.
- Conversely, stakeholder theory says that managers should make decisions that consider the interests of all stakeholders. ESG considerations further complicate this.

Better to use stakeholder theory?

Managers consider the interests of all stakeholders:

- Customers.
- Lenders.
- Employees.
- Suppliers.
- Government and regulatory bodies.
- Communities, both local and globally (including ESG).
- Shareholders/owners.

This was demonstrated by Goldman Sachs (possibly unintentionally) in a research piece on gene therapy last week. It has attracted some [attention](#) from the tech-ier parts of the [blogosphere](#) because of the following section:

Is curing patients a sustainable business model? The potential to deliver “one shot cures” is one of the most attractive aspects of gene therapy, genetically-engineered cell therapy and gene editing. However, such treatments offer a very different outlook with regard to recurring revenue versus chronic therapies, particularly in certain diseases where it is possible to cure a large proportion of the prevalent patient pool (or at least prevent an additional dose from being required for an extended period). While this proposition carries tremendous value for patients and society, it could represent a challenge for genome medicine developers looking for sustained cash flow. GILD is a case in point, where the success of its hepatitis C franchise has gradually exhausted the available pool of treatable patients.

That is a fair point! It also illuminates the conflicting incentives that arise when a country relies on the profit motive to manage public health. There could be all sorts of

Financial Times, April 16, 2018, “Capitalist institution makes strong case for socialised medicine”

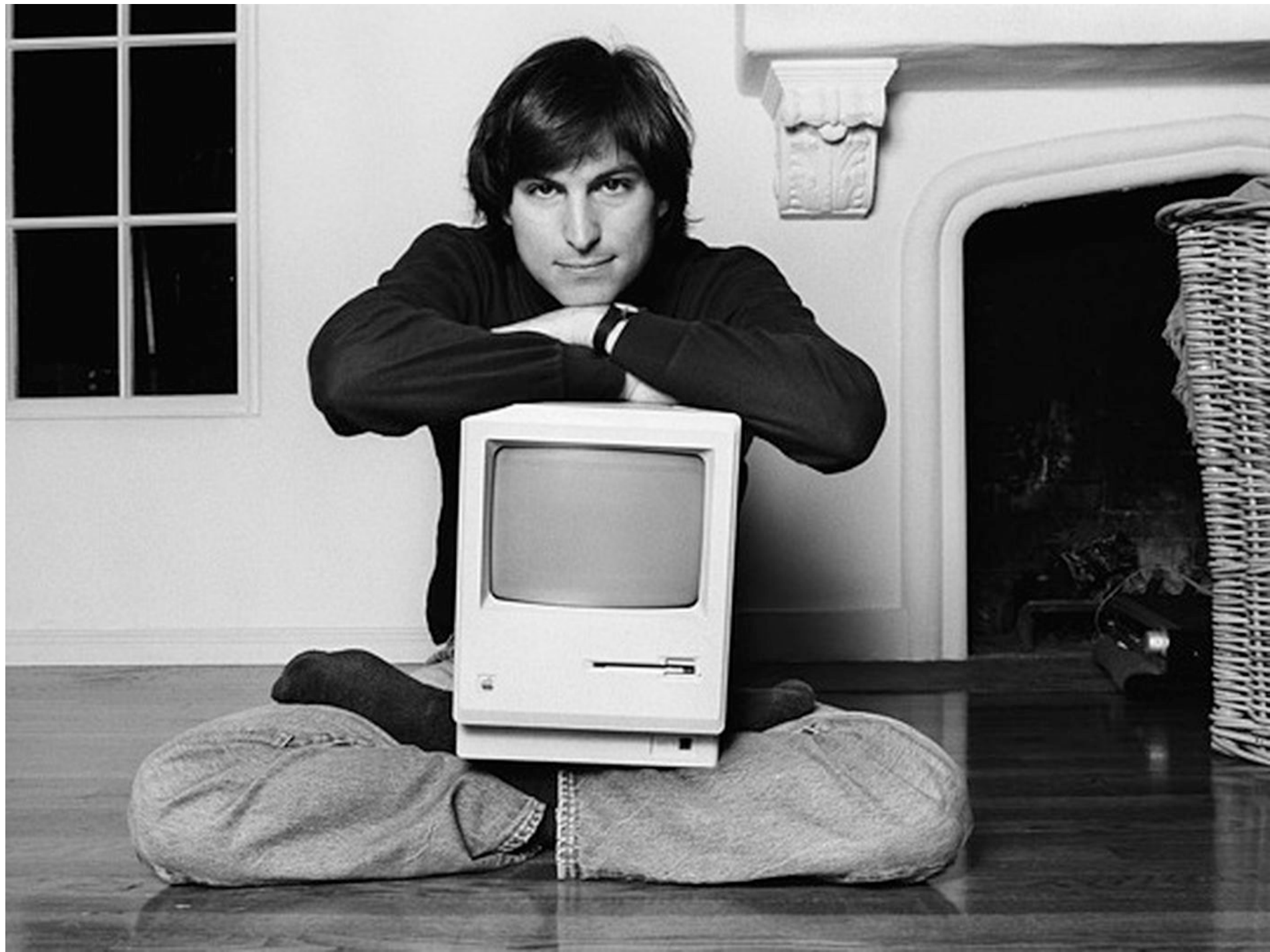
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Managers of public companies face pressures to focus on shareholder value maximization:

Saul Martinez UBS Investment Bank, Research Division - MD & Analyst

... I guess I'll just follow up with sort of a bigger picture type of question. And you guys have a lot of stakeholders, as any bank or any company, employees, the communities you serve, your clients, your shareholders. You obviously want to be seen as a good corporate citizen whose actions are doing good for society at large. But sort of in the grand scheme of things, I mean how important is generating an 11% to 13% ROE for shareholders? And the question isn't meant to be confrontational or grandstanding in any ways, but it does feel like it's sort of in the background of discussions you have with investors, is Wells going to pull the cost leverage part if they can, are they going to communicate that? Are they going to be more apt to change pricing or increase deposit pricing or whatever it is. But I kind of wanted to just ask sort of how do you think about the push and pull of how you run the business and decide how to address different stakeholders' interest?

Timothy J. Sloan Wells Fargo & Company - CEO, President & Director Yes. I'll start, and I'm sure John will want to jump in, but we appreciate that we have a lot of stakeholders, and it is not lost on us, that one of our most important stakeholders are you, our shareholders. And we have historically provided exceptional returns on an absolute and a relative basis. I think our returns are -- continue to be very, very good. Others have caught up, we appreciate that. But make no mistake, we understand that increasing our capital return to our shareholders, like we did to this quarter, when we were up to \$4 billion and continuing to grow ROE, is very important. And that's one of the reasons why then we introduced our 6 aspirational goals to our team, that we were very clear that providing the best long-term returns to our shareholders in the industry is one of our goals, and we're going to achieve that goal.

Source: WFC earnings call, 3Q17 10.13.17

There are a lot of different ways to think about how to set organizational objectives, many of which conflict and involve tradeoffs, and figuring out which to follow is not obvious or easy.

Without establishing specific objectives (as well as a way of managing conflicts between them), it is hard to see how an organization can succeed.

The need for management to be accountable for ESG (and other non-financial) goals further complicates this.

ROIC is the key financial metric:

$$ROIC = \frac{NOPAT}{Invested\ Capital}$$

NOPAT = Earnings from core operations net of tax.

Invested capital = Investment in core operations (PP&E, intangibles, working capital).

ROIC > opportunity cost of capital (WACC)
→ value created

Sources of Competitive Advantage

Price premium

Innovative products: Difficult-to-copy or patented products, services, or technologies

Quality: Customers willing to pay a premium for a real or perceived difference in quality over and above competing products or services

Brand: Customers willing to pay a premium based on brand, even if there is no clear quality difference

Customer lock-in: Customers unwilling or unable to replace a product or service they use with a competing product or service

Rational price discipline: Lower bound on prices established by large industry leaders through price signaling or capacity management

Cost and capital efficiency

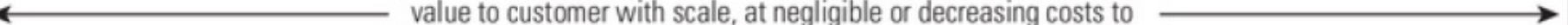
Innovative business method: Difficult-to-copy business method that contrasts with established industry practice

Unique resources: Advantage resulting from inherent geological characteristics or unique access to raw material(s)

Economies of scale: Efficient scale or size for the relevant market

Scalable product/process: Ability to add customers and capacity at negligible marginal cost

Network effects: Products or services that offer increasing value to customer with scale, at negligible or decreasing costs to scale, possibly combined with customer switching costs



The Rules of the Game

Agency problems in large modern corporations

Because shareholders delegate decision-making authority to managers whose interests are not fully aligned with those of shareholders, there are *conflicts of interest between managers and shareholders*.

- Managers may “shirk,” over-consume “perks,” or even expropriate shareholder wealth for their own purposes.
- Economists refer to this as *moral hazard*.
- These types of conflicts of interests are the crux of **corporate governance**.

The \$65 Million Perk for CEOs: Personal Use of the Corporate Jet Has Soared

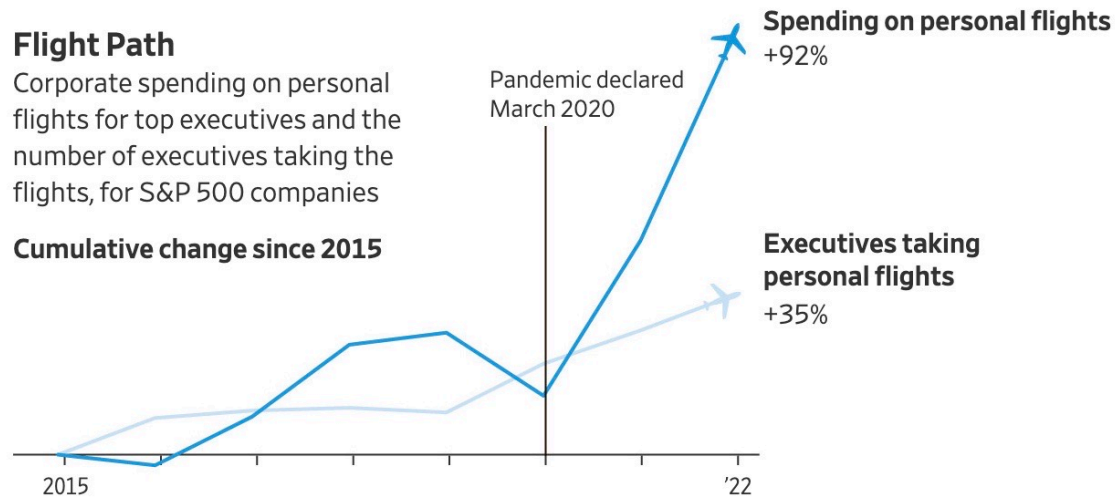
Company spending on the benefit has climbed 50% since before the pandemic



Flight Path

Corporate spending on personal flights for top executives and the number of executives taking the flights, for S&P 500 companies

Cumulative change since 2015



Source: Equilar

Source: WSJ, January 16, 2024.

Incentive problems in organizations

The principal-agent (**moral hazard**) problem between top managers and shareholders in large public companies is an example of a more general problem:

→ How do the organization's leaders ensure their teams take actions consistent with the organization's objective(s)?

- Team members may not have the right incentives and may shirk or otherwise act opportunistically.
- Team members may not understand overall firm strategy/objectives in a way they can execute on.

Rules of the Game: How We Achieve our Objectives

Internal “rules of the game”:

1. A system for **assigning decision rights** to those throughout the organization.
2. A **control system** that:
 - Measures and evaluates performance.
 - Provides rewards and punishment.

What does this help achieve? **Coordination** and **Communication**, which are critical for organizational success.

When you get this wrong:

- Rio Tinto's destruction of a 46,000 year-old Aboriginal site in Western Australia (to expand an iron ore mine with incremental value of \$115M).
- CEO had not read archaeological report that would have alerted C-suite to value of site.
- Boeing's 737 MAX failure – Poor governance at board level; pressure to develop new product too quickly (to compete with A320Neo); failure to consider opportunity costs of 787 development.
- Communication/coordination failures that led to near-successful assassination attempt on President Trump.

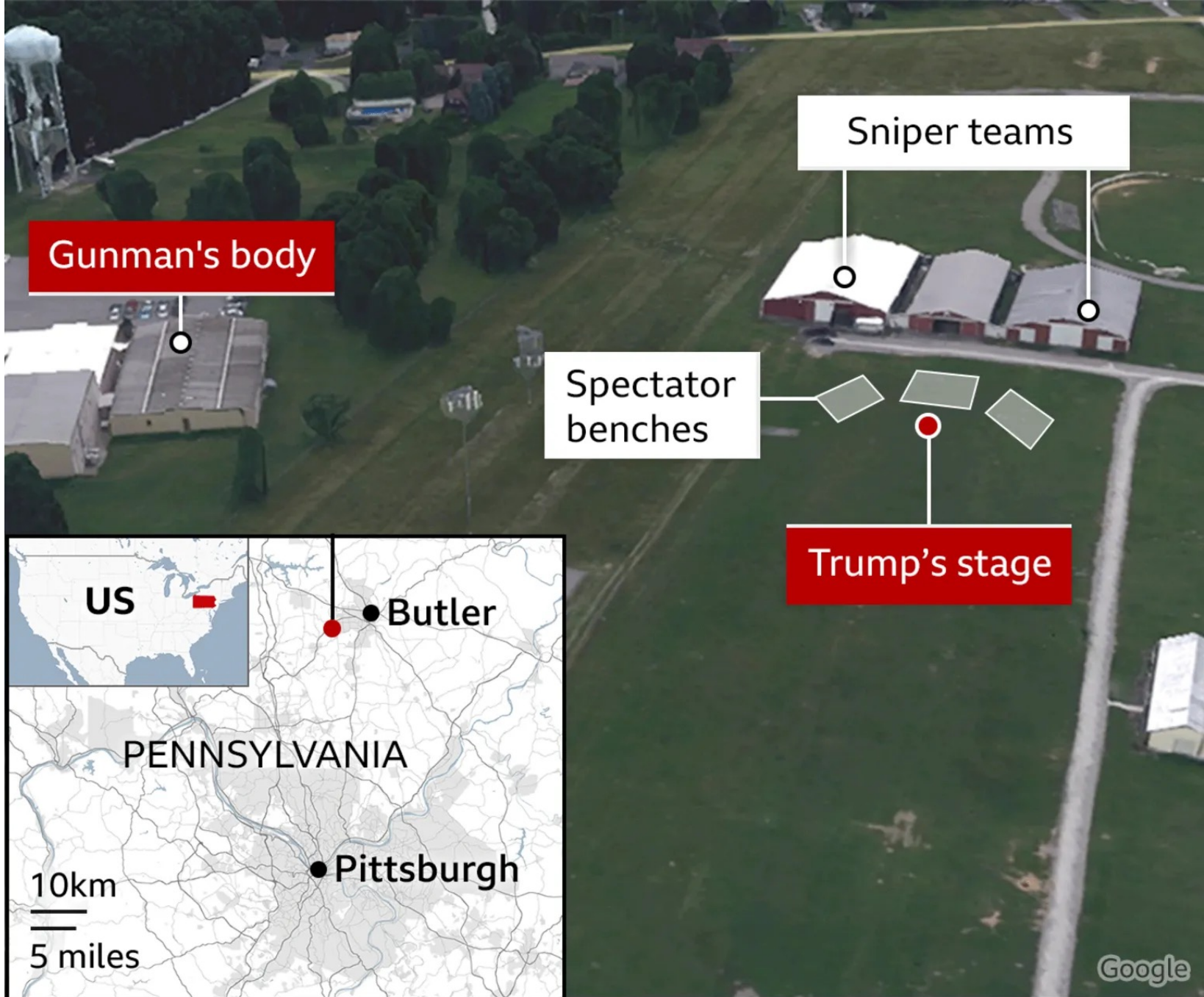


Table 1

Comparison of the three control techniques analysed using the performance management framework

Question	Budgetary control	EVA TM	Balanced scorecard
1. Objectives	Financial objectives: –profit; –cash flow; and –ROCE	Single financial objective.	Multiple objectives based on strategy.
2. Strategies and plans	Means/end relationships not formally considered, although budget is based on a plan of action.	Delegated to responsible managers. May be considered when setting targets.	Implicit in selecting some performance measures; no formal procedures suggested.
3. Targets	Best estimates for financial planning; literature on target-setting gives some guidelines for control.	Some guidance is given with respect to ‘inheritance effect’.	Not considered, despite being central to ‘balance’.
4. Rewards	Not addressed, despite many rewards now being made contingent upon budget achievement.	Appropriate incentive schemes a central part of the methodology.	Not addressed.
5. Feedback	Short-term feedback of budget variances. Incremental budgeting from year to year.	Some discussion of longer-term impact.	Reporting of performance assumed, but no explicit guidance given.

Source: David Otley, “Performance management: A framework for management control systems research,” *Management Accounting Research* 10 (1999): 363-382.

Behavioral aspects of managerial accounting

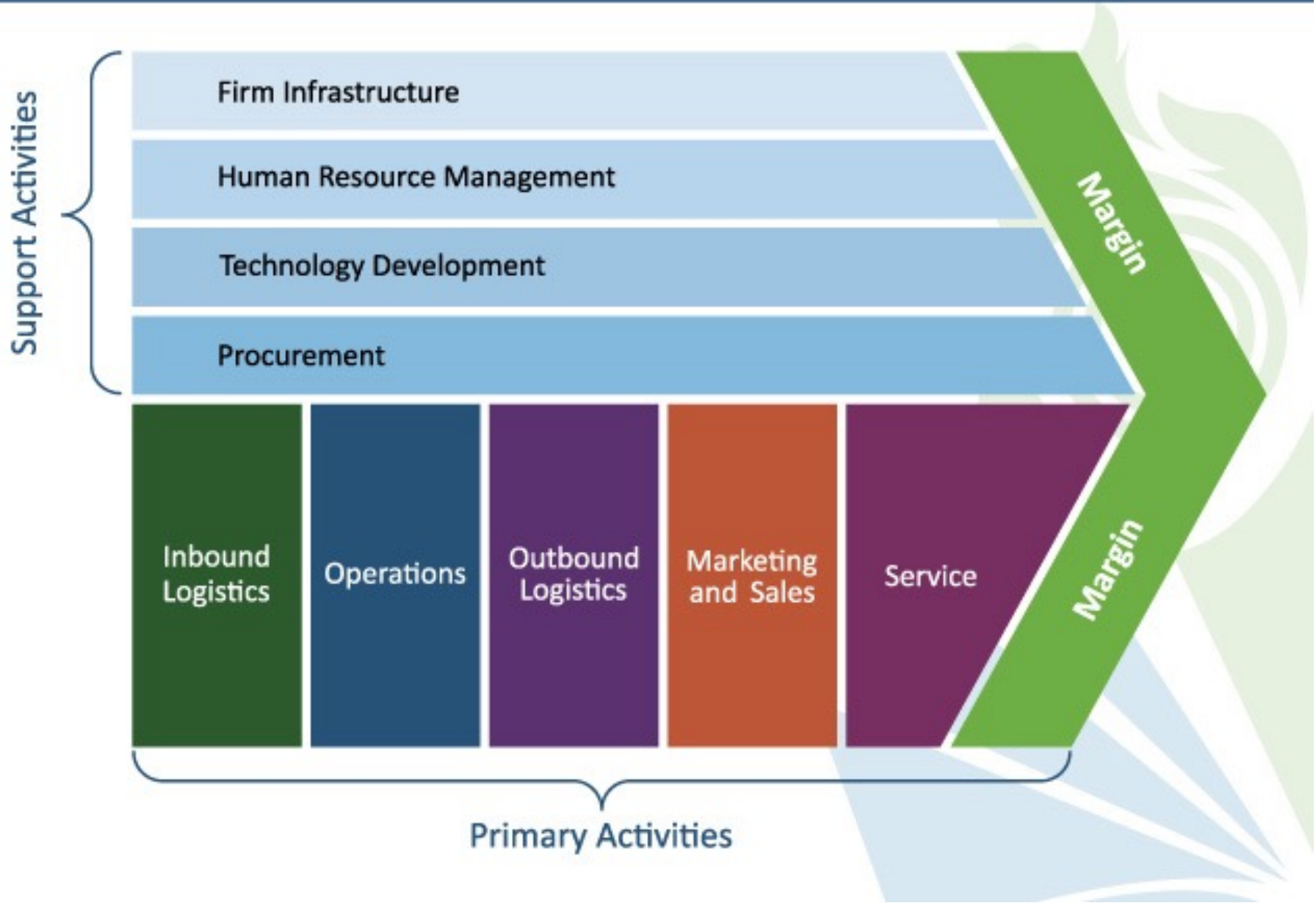
While much of our focus is on costs and cost data, managerial accounting also has to do with psychology and human behavior:

- *Rules of the Game* will not be effective unless everyone buys in.
- Having business unit managers *participate* in setting goals, targets, and budgets works better than when budgets are *imposed* on a top-down basis.
- Should we set “stretch” targets? Do they motivate or de-motivate employees?

Costs and Cost-Volume-Profit

- Organizations undertake activities that create value for customers or other stakeholders.
- **Value chain:** Activities that increase the value of the goods or services sold to customers (i.e., these add customer utility to the goods and services).
- **Activities consume resources and so cause costs.**
Managers seek to identify activities that don't add value and eliminate those activities and the associated costs.
- Moving inventory from the factory to a warehouse is costly but may not add value.

Value Chain Analysis



- **Cost drivers:** underlying factors (variables) that cause costs. Understanding cost drivers allows us to control and manage costs.
- In Booth Executive Education, the number and variety of programs (custom v. open), the number of locations (Gleacher, London, Hong Kong, others), the number of instructors, and the number of participants enrolled all drive costs.
- The number of participants drives catering and printing costs; the number of locations drives faculty and staff travel costs as well as adding complexity to logistics related to program delivery.

Our interest is in **costs**, not **expenses**.

- **Expenses** are defined in financial accounting (GAAP) as the cost of resources consumed in a period to generate revenue. GAAP rules determine when costs are expensed (e.g., rules for depreciation, inventory costs/COGS, R&D spending).
- **Costs** are either actual (**cash**) **outlays** or **opportunity costs**. Outlays are usually cash outflows (e.g., the cost of raw materials purchases; wages paid to employees) that occur now or later. Accounting systems typically only capture actual outlays.

Are dogs better off in a world with mobile phones?



- The **opportunity cost** of a resource is the value of the resource in its most highly-valued *alternative* use (i.e., next-best use).
- Let's say we decide to use existing land, purchased 20 years ago, to build a new production facility. What is the opportunity cost (for NPV calculation)?
 - GAAP measures the **historic cost** of the land: cash outlay when land purchased. Not a good measure.
 - The land's current **net realizable value** (expected sales price net of transactions costs) is a better measure.
- Measuring opportunity cost depends on the **opportunity set**—the set of alternative uses of the resource.

- The amount spent on the land 20 years ago is a **sunk cost** incurred in past (cannot be changed).

Sunk costs are unavoidable and non-recoverable.

- The cost to Airbus of developing the A380, at around \$25B, is sunk – unavoidable and non-recoverable.
- Sunk costs are **irrelevant for decisions** made today (because today's decisions are inherently forward looking).
- However, sunk costs may be relevant for other purposes – knowing how much a managers spent in the past may be useful for **control** (to hold managers responsible for past expenditures, such as M&A).

Last updated: January 17, 2013 8:41 pm

Rio chief quits after \$14bn writedown

By Neil Hume in Sydney and Helen Thomas and David Oakley in London



Sam Walsh, head of the iron ore division, will become chief executive of Rio Tinto

Tom Albanese has paid the price for failures at Rio Tinto, stepping down as chief executive following \$14bn in writedowns at the Anglo-Australian group's aluminium and coal businesses.

Rio said on Thursday it would write down the value of its aluminium assets by \$10bn-\$11bn, mostly related to its disastrous \$38bn purchase of Canada's Alcan in 2007. The group will also take a \$3bn charge over coal assets in Mozambique, purchased only two years ago for \$3.7bn.

Mr Albanese will be replaced by Sam Walsh, the head of Rio's iron ore division – its biggest and most profitable business.



Click to enlarge

Under Mr Albanese's stewardship in late 2007 Rio rejected a \$67bn takeover approach from BHP. Eighteen months later it launched a \$15bn rights issue instead of accepting a cash injection from Chinese aluminium producer Chinalco.

Rio's chief executive is the latest senior departure in the mining sector, after Cynthia Carroll last year said she would leave [Anglo American](#) and Mick Davis agreed to stand aside at [Xstrata](#) to help seal the group's merger with commodities trader [Glencore](#).

The board of [BHP Billiton](#), the largest global miner, has also started looking for a successor for chief executive Marius Kloppers.

While analysts expected further writedowns over the beleaguered aluminium operations, Jan du Plessis, Rio's chairman, called the Mozambique impairment "unacceptable". Doug Ritchie, the executive who oversaw the deal, will also leave the group.

The pair will receive no lump sum payments on their departure or performance awards for this

Cost of past investments is sunk but can be used to hold managers accountable for poor decisions

Fixed costs, variable costs, and the relevant range.

- An operations manager at American Airlines wants to know how costs will decline if the number of weekday flights from Chicago O'Hare to New York LaGuardia is cut from 12 to 6 – *an important question in March 2020.*

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- To answer this, we need to know what costs are variable and what costs are fixed, *within the relevant range of activity*.
- A related but different question is the extent to which these costs are *avoidable* in a cash flow sense.

Variable costs: change in direct proportion to the level of activity.

- Fuel costs are approximately proportional to the number of flight-miles, and so are variable. Crew costs depend on number of flights (their time is costed on an hourly basis) and so are also variable but with respect to flights (or flight-hours) a different activity measure.

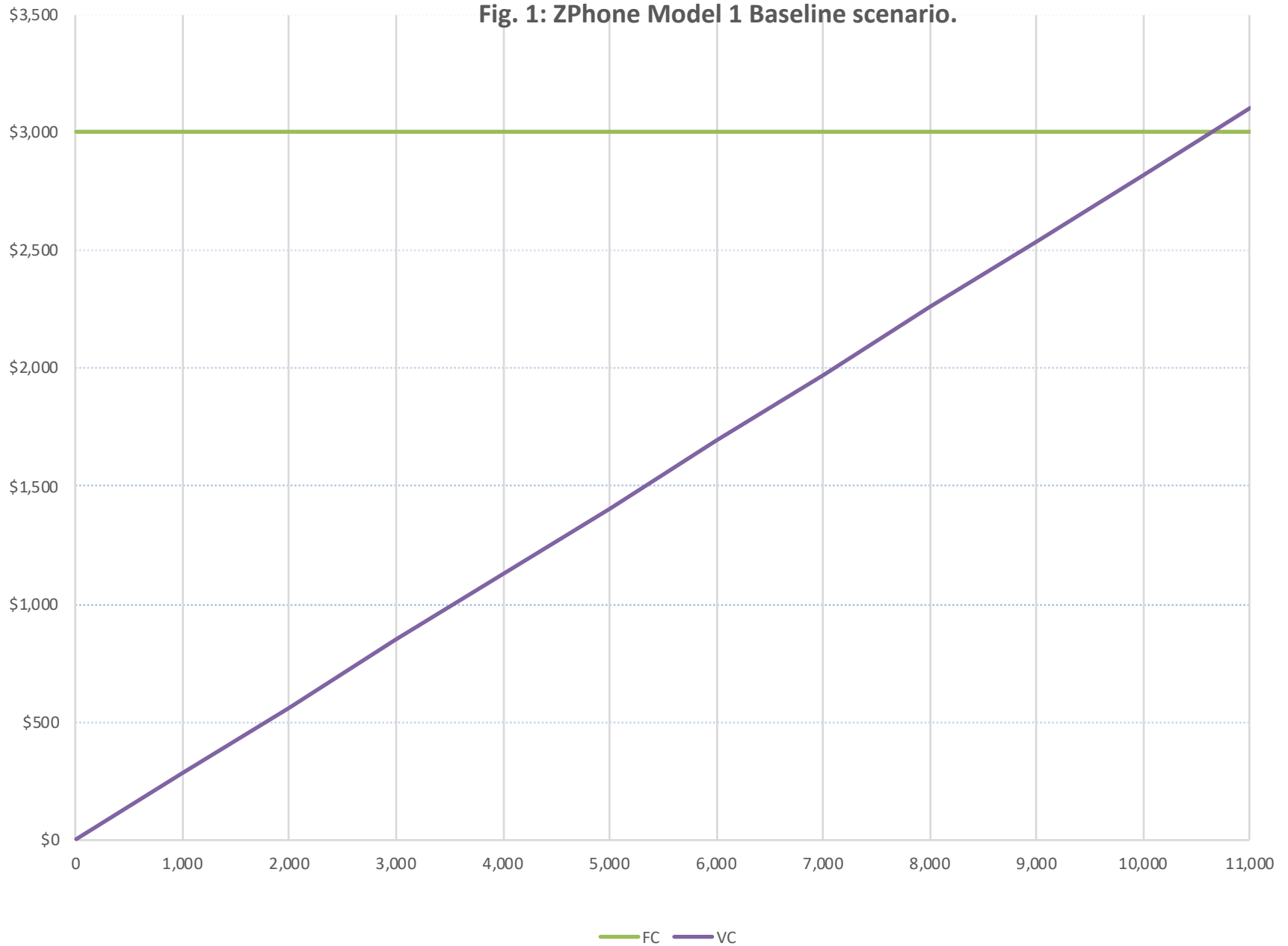
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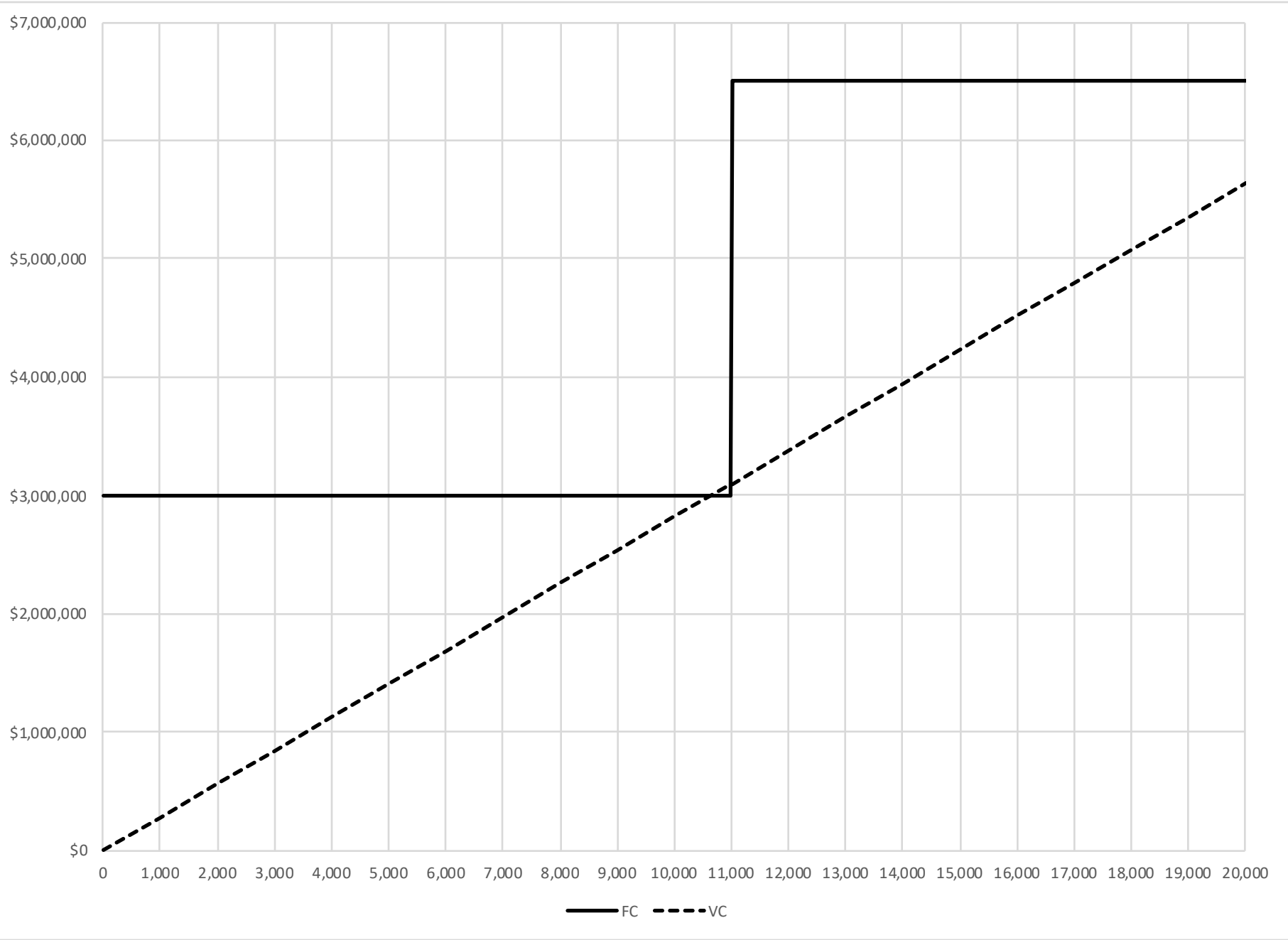
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- Costs can only be variable with respect to a specific *cost driver* (*activity measure*).
- Defining the right cost driver is critical. For airlines, is this number of passengers, number of flights, number of flight-miles, number of flight-hours, or something else? If we're interested in fuel costs, it depends on what best explains fuel consumption.

Fig. 1: ZPhone Model 1 Baseline scenario.



- **Fixed costs:** do not change in this range of activity. If we continue to operate flights to LGA from 6am-10pm each day, the cost of gate agents at LGA is fixed because we employ the same number of agents for the same hours regardless of the number of flights. The cost of landing “slots” at LGA is also fixed, as are equipment costs such as aircraft depreciation and lease rental payments.
- Over a longer period, fixed costs become variable: if we reduce the number of flights we can employ fewer gate agents each day and reduce labor costs. However, these costs may not be **avoidable** in a cash flow sense (we may still have to pay them).

-
- Some costs are a combination of fixed and variable costs, including **step costs** and **mixed/semivariable** costs.
 - The cost of cleaning crews is a step cost if one crew can clean 7 aircraft in a four-hour shift (an additional crew is needed if there are more than 7 flights in this period).



	American Airlines (AAL)								
	3Q20	10-Q							
	11.3.20		3Q20	%	% change		3Q19	%	
	Operating revenues:								
	Passenger		2,540		-76.9%		10,995		
	Cargo		207		-0.5%		208		
	Other		426		-39.8%		708		
	Total operating revenues		3,173		-73.4%		11,911		
	Operating expenses:								
	Aircraft fuel and related taxes		453	17.8%	-77.2%		1,989	18.1%	
	Salaries, wages and benefits		2,705	106.5%	-16.0%		3,219	29.3%	
	Regional expenses		914	36.0%	-52.7%		1,933	17.6%	
	Maintenance, materials and repairs		337	13.3%	-44.8%		610	5.5%	
	Other rent and landing fees		367	14.4%	-30.8%		530	4.8%	
	Aircraft rent		336	13.2%	0.3%		335	3.0%	
	Selling expenses		70	2.8%	-83.5%		424	3.9%	
	Depreciation and amortization		498	19.6%	-0.2%		499	4.5%	
	Special items, net		-295	-11.6%			228	2.1%	
	Other		659	25.9%			1,336	12.2%	
	Total operating expenses		6,044	2.380	-45.6%		11,103	101.0%	
	Operating income (loss)		-2,871				808		

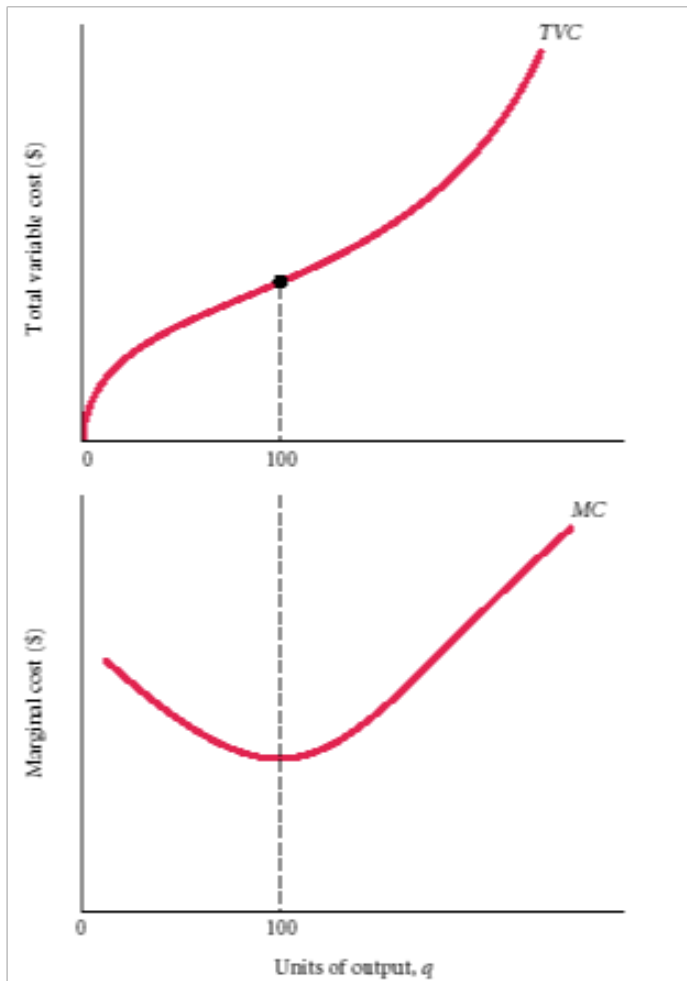
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	Total operating expenses		6,044	2.380	-45.6%		11,103	101.0%		
	Operating income (loss)		-2,871				808			

- Just because costs are variable or fixed doesn't necessarily mean that they are **avoidable** (in the case of variable costs) or **unavoidable** (in the case of fixed costs).
- If we can't layoff employees who assemble products (direct, variable costs) those costs are not avoidable.
- Conversely, fixed costs may be avoidable. If we stop using a plant we may be able to avoid utilities costs (like electricity, heating) that are fixed over the range of activity relevant for that plant.
- Key idea: whether costs are **variable** or **fixed** does not say much about whether the costs are **avoidable** in a cash flow sense, which may be our key concern.

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- **Marginal costs:** the *incremental* cost of each additional unit of activity. These are unlikely to be constant over the relevant range of activity.



Common cost structure (i.e., production function):

Marginal costs are high at first, then decline, and then increase again.

Marginal cost: slope of the total cost curve at each point.

Variable costs: can approximate as the slope of a straight line fitted to the total cost curve where it is approximately linear (the relevant range).

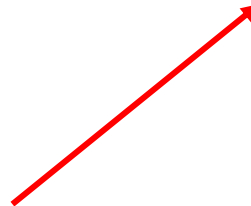
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- **Cost drivers:** underlying factors (variables) that cause costs. Understanding cost drivers allows us to control and manage costs.
 - What are the cost drivers at your local Starbucks store?

The Starbucks Turnaround That Has Baristas and Customers Steamed

Improved service, better seating and heartfelt messages are at the heart of Brian Niccol's turnaround plan, but baristas say they're under pressure.



The Strawberry Matcha Strato Frappuccino, center, at Starbucks requires six ingredients and two blenders to make. Nico Schinco for The New York Times



- **Cost drivers:** underlying factors (variables) that cause costs. Understanding cost drivers allows us to control and manage costs.
- What are the cost drivers at your local Starbucks store?
 - Number of customers.
 - Number of customers per unit time.
 - Variety of products (beverages, food).
 - Delivery mode: in-person, on-line orders, drive-thru.
 - Local demographics/location/customer type.

Contribution Margin and Cost-Volume-Profit.

Contribution margin = Revenue – Variable Costs.

Why important? Tells us how much \$\$ are available to cover all non-variable costs and deliver profit (and cash flow).

$$CM = (p - vc).y \quad \text{Total \$ of Contribution Margin}$$

p = unit price

vc = variable cost per unit

y = volume in units

(We assume p and vc are constant over a defined relevant range.)

$$\begin{aligned}\text{Total Profit } \$ &= (p - vc).y - FC \\ &= cm.y - FC\end{aligned}$$

where cm = \$ contribution margin per unit.

This means **breakeven** is:

$$y^* = FC/cm$$

Once we sell y^* units, we cover our fixed costs – every incremental unit contributes an \$ cm to the bottom line.

Contribution Margin example:

Consider an upcoming Exec Ed program at our London campus.

We charge \$8,000 per participant for the five-day program.

We incur costs of:

- \$1,000 per participant (catering, materials)
- \$39,000 for faculty compensation and travel.
- We use two London team members to handle logistics and make use of the facility itself (70-seat classroom, breakout rooms, and meal/break spaces).

We will decide two weeks in advance whether the program will run based on number of participants who have registered.

- Number of participants is the activity measure.
- $vc = \$1,000$ and direct $FC = \$39,000$.

$$\rightarrow cm = \$8,000 - \$1,000 = \$7,000$$

→ Breakeven is:

$$y^* = \$39K / \$7K = 5.57 \text{ participants.}$$

i.e., with **six** participants total $CM = \$42K$ and we cover our fixed costs.

-
- If we have 10 participants, we feel pretty good: total CM of \$70K easily covers fixed costs of \$39K.
 - If we're at 5 participants, we cancel to avoid incurring fixed costs of \$39K (CM = \$35K).

- If we have 10 participants, we feel pretty good: CM of \$70K covers fixed costs of \$39K.
- If we're at 5 participants, we cancel to avoid incurring faculty-related fixed costs of \$39K (CM = \$35K).
- Would this change if the fixed costs included \$9K of *non-refundable* airfare and hotel?

Relevant/avoidable fixed costs now \$30K (i.e., \$9K is sunk), so breakeven is now 5 participants.

This analysis accounts for direct costs, both fixed and variable, in the relevant range of activity (1 to 70 participants, the capacity of the classroom).

Did we miss anything?

→ **Opportunity cost** of two local staff needed to help run the program as well as of the space (what if we can rent out the space for conferences/events?).

Let's say we've decided to run the program.

What if a corporate client comes along at the last minute and wants to send five people but only pay \$5,500 for each?

- We should accept this offer → adds \$22.5K [$5 \times (5,500 - 1,000)$] in total contribution which drops directly to the bottom-line.

Key rules:

If capacity is available, accept additional orders if the unit CM is positive since this will increase bottom-line

- this is how Airbus should have thought about year-by-year decisions to produce new A380s once (sunk) development and production set-up costs had been incurred.

If capacity is constrained and we have multiple products, need to allocate capacity to products that have largest CM per unit of the constrained resource (see ZPhone example).

Satya Nadella, '97

The Microsoft CEO talks about empathy and hard-won hard skills with dean **Sunil Kumar**.

I will start with this: What's your vision for Microsoft?

Having grown up in the company, what is it 23 years or so, I always come back to the very first thing Microsoft did, the BASIC interpreter. It was this notion of empowering others to do great things.

And how do you do that as a business?

I also think, in Chicago style, we're all victims of our economic model. In our case it's about your product. We want to do the best job of making the tools you use. It could be a student writing a term paper or a developer writing an app. My vision is to take that productivity and empowerment of individuals and organizations and express it in this new world, which is mobile-first, cloud-first. The only way you're going to have that mobility of the human experience—not the device—is by having cloud orchestrate the movement.

Regarding that Chicago-style thinking, is there a Booth class that was particularly memorable?

It was a class that I took with [now-retired] professor **Marvin Zonis** on leadership. He stressed that EQ trumps IQ in the long run—the E is for empathy.

How did you carry this into your career at Microsoft?

When I've had my failures—when I've been impatient or not listened to someone—it was because I didn't enlist my sense of empathy. Empathy helps me understand what others expect, what they need to succeed, and what the authentic ways are to communicate. We're all social beings.

Satya Nadella '97, is a graduate of the Weekend MBA Program. For two years, he commuted to Chicago from his job at Microsoft headquarters in Redmond, Washington, where he was an engineer and program manager working on Windows. He was named CEO of the software giant in February 2014.

What about the hard-skills classes?

Cost Accounting was painful. But I tell you, cost accounting is everything. When we think about a new storage system or the amount of bits that we are moving into a data center, it comes down to the cost of photons.

Managerial Accounting was perhaps the best class. That's because the biggest mistakes we make lie in not knowing which are variable and which are fixed costs. That's especially true as we move to the cloud. You can make all kinds of pricing mistakes.

Is there a role for nontechnical MBAs at tech companies?

It takes all functions. If you're an expert in finance or marketing, there are fantastic opportunities. At the end of the day as a tech company, it's about the innovation. It's about the product. And in our case it's about computer science. Of course you need the engineers. But it's really the interdisciplinary work that is going to create the magic.

What magic comes next?

There is a real possibility now of superintelligence—in 20 years, 50 years, but it's going to happen. What it is going to do to humanity and human evolution is fascinating. I'm an optimist, because we in the technology industry have to be. We're going to get more out of our time. Technology has always led us to have better lives. ♦

